



**MALLA REDDY TECHNICAL CAMPUS**  
A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH  
*Deemed to be University*  
Approved by UGC & AICTE New Delhi  
Maisammaguda, Medchal (Dist), Hyderabad - 500100, Telangana



# PROJECT REPORT

# SCHOLARSHIP CANDIDATE SELECTION SYSTEM

Using Greedy Algorithms, Sorting and Dynamic Programming

## DAA HACKATHON PROJECT

Submitted By

Team Members:

S. AASHRITHA

A.LAVANYA

M.SAI MANOGNA

WAHEDA ZEENATH

Department: Artificial Intelligence and Machine Learning

Academic Year: 2026–2027

Scholarship Candidate Selection System | DAA Project

# 1. Abstract & Introduction

## Abstract

The Scholarship Candidate Selection System is a web-based application designed to simplify and automate the process of selecting candidates for scholarships. The system evaluates candidates using academic performance, financial need, and eligibility constraints. It applies sorting, greedy selection, and dynamic programming concepts to rank candidates and allocate limited scholarship resources according to predefined rules.

The project demonstrates how Data Structures and Algorithms can be applied to a practical resource-allocation problem. By automating candidate evaluation and ranking, the system can reduce manual effort and provide a consistent, structured workflow for scholarship selection.

## Introduction

Scholarship programs provide financial assistance to students who satisfy academic, financial, or other eligibility requirements. When the number of applications is large, manually reviewing and comparing candidates can take considerable time and effort.

The proposed system provides a computerized approach in which candidate information is collected, eligibility is checked, candidates are ranked using defined criteria, and suitable candidates are selected under scholarship constraints. The project connects theoretical DAA concepts with a practical application.

# 2. Problem Statement, Objectives & Scope

## Problem Statement

Selecting scholarship candidates from a large applicant pool requires consideration of multiple factors such as academic marks, financial need, eligibility conditions, and limited scholarship availability. A manual process can be time-consuming and difficult to scale. The proposed system aims to evaluate applicants systematically, rank eligible candidates, and select candidates while respecting predefined constraints.

## Objectives

Automate the scholarship candidate selection process.

Check candidate eligibility efficiently.

Rank eligible candidates using defined criteria.

Consider academic performance and financial need.

Allocate limited scholarship resources using algorithmic methods.

Reduce manual processing and comparison effort.

Provide clear and structured selection results.

## Scope

The system can be used by colleges, educational institutions, scholarship organizations, and similar organizations that need to select beneficiaries from multiple applications. It can be extended to support different scholarship schemes, customized eligibility rules, multiple categories, and different budget limits.

# 3. Features & Functional Requirements

## Features

**Candidate Registration:** Collects candidate details such as name, marks, financial information, and eligibility data.

**Eligibility Verification:** Checks whether a candidate satisfies the defined eligibility conditions.

**Candidate Ranking:** Ranks eligible candidates using the configured scoring criteria.

**Scholarship Selection:** Selects candidates while considering scholarship availability or budget constraints.

**Automated Processing:** Reduces repetitive manual calculations and comparisons.

**Results Display:** Shows rankings, eligibility status, and selected candidates clearly.

## Functional Requirements

The system shall accept candidate information.

The system shall validate candidate inputs.

The system shall check eligibility conditions.

The system shall calculate a candidate score or priority.

The system shall sort eligible candidates.

The system shall apply selection constraints.

The system shall display selected and non-selected candidates.

The system shall present understandable results to the user.

## 4. Algorithms & Methodology

### Algorithms Used

#### Sorting

Sorting is used to arrange eligible candidates according to their calculated score or priority. Once candidates are sorted, higher-priority applicants can be processed first. The exact sorting technique can be selected based on the implementation requirements.

#### Greedy Algorithm

The Greedy approach makes the best available selection at each step according to the defined priority rules. For example, candidates can be considered from highest priority to lowest priority while ensuring that the available scholarship budget or quota is not exceeded.

General process: Rank Candidates → Select Highest-Priority Eligible Candidate → Check Constraint → Continue Until the Resource Limit Is Reached.

#### Dynamic Programming

Dynamic Programming can be used when scholarship allocation involves a limited budget and different scholarship amounts. The problem can be modeled as an optimization problem in which feasible combinations are evaluated to identify an allocation that satisfies the defined constraints.

### Methodology

Candidate Data

Eligibility Verification

Score Calculation

Candidate Ranking

Greedy / Dynamic Programming Selection

Final Selected Candidates

Result Display

## 5. Technologies & Implementation

### Technologies

| Technology / Tool                    | Purpose                                |
|--------------------------------------|----------------------------------------|
| HTML, CSS, JavaScript / React        | Frontend and user interface            |
| Python / Node.js                     | Backend and algorithm implementation   |
| MySQL / MongoDB                      | Candidate and application data storage |
| Sorting, Greedy, Dynamic Programming | Core DAA techniques                    |
| VS Code                              | Development environment                |
| Git / GitHub                         | Version control                        |

### Implementation

**Candidate Module:** Collects candidate details including academic marks, financial information, and eligibility data.

**Eligibility Module:** Checks whether applicants satisfy minimum requirements.

**Ranking Module:** Calculates scores and orders eligible candidates according to the selection criteria.

**Selection Module:** Applies the selected algorithm and resource constraints to identify beneficiaries.

**Result Module:** Displays ranking, eligibility status, selection status, and other relevant information.

## 6. Results, Testing & Performance

### Results

The system produces a ranked list of eligible candidates and identifies candidates who satisfy the defined scholarship selection criteria. The following is sample output; actual website results should be inserted in the final report.

| Candidate   | Marks | Financial Need | Eligibility | Rank | Status       |
|-------------|-------|----------------|-------------|------|--------------|
| Candidate A | 92    | High           | Eligible    | 1    | Selected     |
| Candidate B | 88    | High           | Eligible    | 2    | Selected     |
| Candidate C | 85    | Medium         | Eligible    | 3    | Selected     |
| Candidate D | 78    | Low            | Eligible    | 4    | Not Selected |

## Testing

**Eligible Candidate** – Input: All conditions satisfied. Expected: Candidate considered for selection.

**Ineligible Candidate** – Input: One or more conditions not satisfied. Expected: Candidate excluded.

**Limited Budget** – Input: Eligible candidates exceed available budget. Expected: Selection respects budget constraint.

**Equal Scores** – Input: Two candidates have similar scores. Expected: Secondary criteria applied.

## Performance

The system is designed to process candidate data systematically and reduce manual comparison. Performance depends on the number of candidates, data-processing operations, sorting method, and selection algorithm. Efficient implementations can support larger applicant sets while keeping the selection workflow structured.

## 7. Conclusion, Future Scope & References

### Conclusion

The Scholarship Candidate Selection System demonstrates the practical use of Data Structures and Algorithms in a scholarship allocation problem. By combining eligibility checking, scoring, sorting, greedy selection, and dynamic programming, the system provides a structured method for evaluating and selecting candidates under defined constraints.

The project can reduce repetitive manual effort and provides a foundation for future improvements such as automated verification, advanced optimization, and analytics.

### Future Scope

AI-assisted candidate recommendation.

Automatic document verification.

Student login and application tracking.

Email or SMS notifications.

Support for multiple scholarship categories.

Advanced fairness and coverage constraints.

Cloud deployment and mobile application support.

Administrative analytics and reporting dashboard.

## References

Thomas H. Cormen et al., Introduction to Algorithms, MIT Press.

Python Documentation, Python Programming Language.

React Documentation, React JavaScript Library.

MySQL Documentation, MySQL Database System.

GeeksforGeeks, Data Structures and Algorithms learning resources.

Academic and research resources related to scholarship allocation and optimization.